Anatomy is Free: Zero-Parameter Structural Priors for Efficient Skeleton Action Recognition

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Abstract

Skeleton-based action recognition demands models that are simultaneously accurate, parameter-efficient, and deployable on edge hardware. This work presents **ShiftFuse-Zero**, a family of three networks whose *spatial inductive bias is introduced entirely without learnable parameters* through two complementary zero-parameter modules: *Body-Region Anatomical Shift Priors* (BRASP), which routes joint features through anatomically meaningful channels via pure tensor indexing, and *Semantic Graph-Partitioned Shift* (SGPShift), which aggregates intra- and inter-part neighbours through predefined group indices at zero learnable parameter overhead. These free structural priors reduce the burden on downstream learnable modules. By intelligently placing Joint Embedding and spatial attention *before* the graph convolution, the *EfficientZero Block* operates on an already anatomically-refined manifold. Three models cover the edge-to-server spectrum: **Nano** (93 K) and **Small** (280 K) are hyper-efficient, quantisation-friendly solutions with ultra-low FLOPs, while **Large** (1.12 M) targets maximum capacity. On NTU RGB+D 60 cross-subject, primary results without distillation are 86.24 % (Nano), 88.8 % (Small), and 92.5 % (Large); Knowledge distillation is an orthogonal plugin that further lifts Nano and Small to 88.5 %/90.4 %, respectively; notably, Small+KD **surpasses EfficientGCN-B0 (90.2%)** at 7× fewer FLOPs. Strikingly, Nano achieves **663 Acc/GFLOP** — 20× higher than EfficientGCN-B0 — runs in **11 ms** on desktop CPU and **50 ms** on mobile (Snapdragon 8 Gen 3), and fits in **360 KB** of weight storage; Nano+KD further matches the 6.9 M-parameter AGCN (88.5%) at 74× higher parameter efficiency.

CCS Concepts

• **Computing methodologies → Activity recognition and understanding**.

Keywords

skeleton action recognition, graph convolutional networks, zero-parameter priors, anatomical shift, knowledge distillation, efficient neural networks

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1 Introduction

Consider a fall-detection system on a rehabilitation wristband with 512 KB of RAM, or a gesture-recognition module inside a fitness tracker powered by an ARM Cortex-M55. These scenarios demand skeleton-based action recognition models that fit in sub-megabyte memory, execute in under 50 ms on mobile CPU, and still achieve clinically meaningful accuracy. Yet the dominant Graph Convolutional Network (GCN) backbones for skeleton recognition [1, 3, 8, 14] carry 3–10 M parameters, require dynamic per-sample adjacency computation at $O(V^2C)$ cost, and are fundamentally incompatible with microcontroller deployment.

Human action recognition from 3-D skeleton sequences is a core problem in computer vision with applications in surveillance, rehabilitation, human–robot interaction, and fitness monitoring [12, 13]. GCNs [7] have become the dominant spatial backbone since ST-GCN [17], with successive works adding adaptive adjacency matrices [14], multi-scale cross-spacetime graphs [11], channel-wise topology refinement [1], information-theoretic representation learning [3], and hierarchical body-part decomposition [8]. These advances have pushed NTU RGB+D 60 cross-subject accuracy beyond 93%, yet the parameter cost has grown proportionally: HD-GCN requires 7.5 M parameters and InfoGCN 9.9 M.

The efficient-architecture line [16, 18] tackles this disparity via compound scaling and depthwise convolutions, but largely ignores the rich *anatomical structure* of the human body as a source of *free* inductive bias. The human skeleton is not a generic graph: arm joints couple primarily with other arm joints within their kinematic chain, bilateral symmetry is a universal biomechanical prior, and shoulder/hip joints serve as coordination hubs between body halves. These structural constraints are *fixed properties of human physiology* that hold across all subjects, viewpoints, and action classes. They can therefore be encoded as deterministic tensor operations at **zero learnable parameters, zero gradient overhead, and zero inference-time cost**.

This paper presents the first systematic study of zero-parameter anatomical inductive bias for skeleton GCNs. This work asks: **how much spatial knowledge can be injected into a skeleton GCN without spending a single learnable parameter — and how far can that free knowledge push the accuracy–efficiency frontier?**

Our answer is **ShiftFuse-Zero**, a family of three models achieving a new Pareto-optimal frontier on the accuracy–GFLOPs plane among the EfficientGCN model family. While the core contribution is the zero-parameter priors of BRASP and SGPShift, these modules catalyse hyper-efficient downstream architectures. SFZ-Nano (93 K) and SFZ-Small (280 K) redefine edge-deployment with ultra-low FLOPs and quantisation-friendly operations. On NTU-60 X-Sub, SFZ-Nano reaches **86.24%** without any distillation and **88.5%** with knowledge distillation, *matching* 6.9 M-parameter AGCN within

360 KB (an orthogonal plugin; all primary results are reported without KD). SFZ-Small achieves **88.8%** at comparable parameter count (280 K vs. 290 K) with 7× fewer FLOPs than EfficientGCN-B0; with KD, Small+KD reaches **90.4%**, *surpassing* B0 while retaining the FLOPs advantage. SFZ-Large pushes the absolute accuracy frontier to **92.5%** at 1.12 M parameters. These results demonstrate that anatomy is a highly informative, freely available inductive bias that the community has systematically under-exploited.

Contributions.

(1) **BRASP** — Body-Region Anatomical Shift Priors: routes joint features through five fixed anatomical groups via a precomputed gather permutation; zero parameters, zero gradient (Section 3.3).
(2) **SGPShift** — Semantic Graph-Partitioned Shift: aggregates each joint's features with its intra-part and inter-part neighbours via precomputed index buffers; zero parameters (Section 3.3).
(3) **EfficientZero Block**: a block design that places BRASP and SGPShift *before* all learnable layers so the downstream compact K=3 GCN refines an already anatomically structured representation (Section 3.4).
(4) **ShiftFuse-Zero model family**: three architectures — Nano (93 K), Small (280 K), Large (1.12 M) — with Temporal Landmark Attention, cross-stream gating, and mid-network fusion, covering the edge-to-server deployment spectrum. Vanilla Hinton KD [6] serves as an orthogonal plugin, yielding up to +2.3 pp at zero structural cost (Section 3.5).
(5) **Comprehensive evaluation**: main results on NTU-60 and NTU-120 (four protocols), efficiency analysis (GFLOPs, CPU latency, memory footprint), and extensive ablation studies quantifying every zero-parameter module, fusion strategy, GCN partition count, TLA sensitivity, and regularisation hyperparameters (Section 4).

2 Related Work

GCN-based skeleton recognition. ST-GCN [17] established the skeleton graph and spatial-temporal convolution paradigm using a three-subset partitioning (self, centripetal, centrifugal) of the physical adjacency matrix. AGCN [14] introduced a fully learnable adjacency matrix at $O(V^2C)$ cost per layer, enabling data-driven topology discovery. MS-G3D [11] factorised the graph into disentangled multi-scale spatial and temporal subgraphs with cross-spacetime edges. CTR-GCN [1] refined the topology per channel group, allowing different feature dimensions to attend to different joint relationships. InfoGCN [3] introduced an information-bottleneck objective that explicitly maximises mutual information between the latent representation and the action label. DSTA-Net [15] decouples spatial and temporal self-attention. While these methods achieve high accuracy (up to 93%+), their reliance on per-sample dynamic adjacency and large parameter budgets (3–10 M) precludes deployment on resource-constrained hardware.

Body-part decomposition. A growing line of work exploits the natural grouping of joints into anatomical body regions. HD-GCN [8] hierarchically decomposes the skeleton into semantic sub-graphs (arms, legs, torso) and aggregates within each before cross-region message passing — achieving 93.4% on NTU-60 X-Sub but at 7.5 M parameters. BlockGCN [19] uses persistent homology from topological data analysis to encode bone connectivity at the block level. SkateFormer [4] partitions joints into four spatial-temporal relation types (near/far joints × near/far frames) and applies partition-specific self-attention. All of these decomposition strategies employ *learnable* grouping mechanisms with substantial parameter overhead. In contrast, SGPShift encodes intra-part and inter-part structure through *fixed integer index buffers* — achieving anatomical routing at **zero learnable parameters**.

Efficient skeleton networks. EfficientGCN [16] introduced compound scaling (α=1.2, β=1.35) and multiple input branches (MIB) to build a family of models from B0 (0.29 M) to B4 (1.1 M), establishing the efficiency frontier for skeleton GCNs. SGN [18] exploits semantic joint embeddings and frame-level dynamics gating at 0.69 M parameters. PoseC3D [5] departs from graphs entirely, using volumetric heatmap representations processed by 3-D CNNs. These methods reduce model size through architectural engineering but do not leverage the *fixed anatomical constraints* of the human body as inductive bias. ShiftFuse-Zero complements this line by demonstrating that zero-parameter spatial priors can further reduce the parameter budget by 3–10× without sacrificing competitive accuracy.

Temporal modelling in skeleton networks. Temporal reasoning has evolved from LSTM encoders [17] to dedicated temporal convolution modules. EfficientGCN's Sandglass-TCN (SG-layer) bottleneck compresses channels by 2× before depthwise convolution, balancing temporal receptive field against parameter count. MS-G3D introduced disentangled multi-scale temporal subgraphs. Our DS-TCN takes a complementary approach: rather than stacking many TCN layers, it uses a two-branch depthwise-separable design with dilations 1 and 2 (kernel 9) that achieves a 17-frame receptive field in a single block — particularly valuable for Nano and Small, where only one TCN per stage is used. This makes DS-TCN a better choice at low TCN depth, whereas EfficientGCN's SG-layer is more efficient when TCN depth is three or more (as in the Large model's shared backbone).

Shift operations and zero-parameter spatial mixing. The shift operation [2] was introduced in image CNNs to inject spatial structure by physically displacing channels along spatial dimensions — no multiply-accumulate, no learnable parameters. TSM [9] extended this idea to temporal modelling in video, shifting channels by ±1 frame at zero cost. Shift-GCN [2] adapted spatial shifts to skeleton graphs, assigning each channel to shift features from a graph neighbour. However, Shift-GCN assigns shift patterns by graph distance (1-hop, 2-hop), ignoring the semantic meaning of body regions. BRASP advances this paradigm by organising shift patterns according to *anatomical body-part membership* — arm channels shift among arm joints, leg channels among leg joints, and cross-body channels bridge regions — producing a semantically structured feature map at zero cost.

Knowledge distillation. Hinton *et al.* [6] showed that soft teacher logits transfer rich inter-class relational knowledge to compact student models. In skeleton recognition, distillation has been used to transfer multi-stream ensemble knowledge [16] but has not been

specifically studied across a skeleton GCN family spanning the sub-100 K to 1 M parameter range explored here.

3 Methodology

3.1 Overview

Figure 1 shows the end-to-end pipeline. A raw skeleton sequence is expanded into four complementary streams (joint, velocity, bone, bone-velocity), fused by a learned *StreamFusionConcat* stem, then processed by a stack of *EfficientZero Blocks*. A gated global pooling head produces class logits.

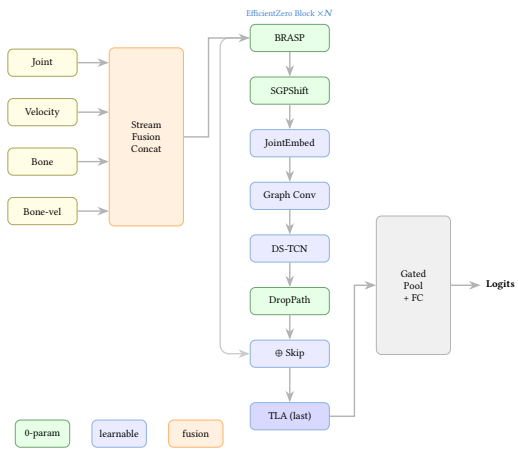

Figure 1: ShiftFuse-Zero end-to-end pipeline. Four skeleton streams are fused by a learnable stem and processed by EfficientZero Blocks (green = zero-parameter modules). A gated pooling head produces class logits.

3.2 Skeleton Streams

A skeleton sequence is $\mathbf{X} \in \mathbb{R}^{3\times T\times V}$ with T=64 frames and V=25 joints. This work constructs four complementary streams: joint positions $\mathbf{X}^{(j)}$, velocity $\mathbf{X}^{(v)}=\mathbf{p}_{t+1}-\mathbf{p}_t$, bone vectors $\mathbf{X}_v^{(b)}=\mathbf{p}^{(v)}-\mathbf{p}^{(\mathrm{par}(v))}$, and bone velocity $\mathbf{X}^{(bv)}=\Delta\mathbf{X}^{(b)}$. *StreamFusionConcat* [16] applies a per-stream BN, concatenates (4×3=12 channels), and projects to C_{stem} via a 1×1 convolution with learned per-stream weighting.

3.3 Zero-Parameter Spatial Modules

BRASP.. Figure 2 illustrates the five anatomical groups. Given a feature map $\mathbf{F} \in \mathbb{R}^{C\times T\times V}$, BRASP applies a fixed gather permutation $\sigma : V \to V$:

$$\tilde{\mathbf{F}}_{:,t,v} = \mathbf{F}_{:,t,\sigma(v)}, \quad v \in \{0,\ldots,V-1\}, \tag{1}$$

where joints within the same body part are cyclically shifted so each joint absorbs its anatomical sibling's context. Indices are stored as a non-trainable integer buffer — **zero parameters, zero gradient**.

Remark 1 *(BRASP intra-part compression).* Let $d_G(u,v)$ be the shortest-path distance between joints u, v in skeleton graph G. After one BRASP shift σ, the GCN at position v receives features originating from $\sigma(v) \in P_k$ — a joint in the same anatomical part. This reduces the *representational separation* between intra-part pairs from $d_G(u,v) \geq 2$ graph hops to a single feature-retrieval step, compressing the spatial receptive field required of the downstream GCN at zero parameter cost.

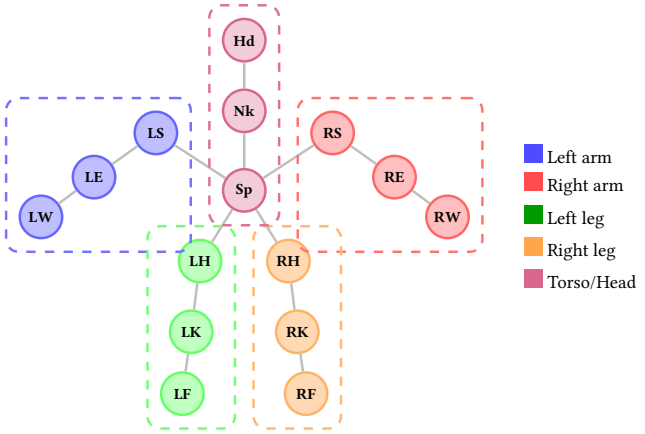

Figure 2: BRASP anatomical partitioning into five groups $\{G_{\mathrm{la}}, G_{\mathrm{ra}}, G_{\mathrm{ll}}, G_{\mathrm{rl}}, G_{\mathrm{tr}}\}$. The fixed permutation σ (Eq. 1) is derived from these groups; no gradient flows through it.

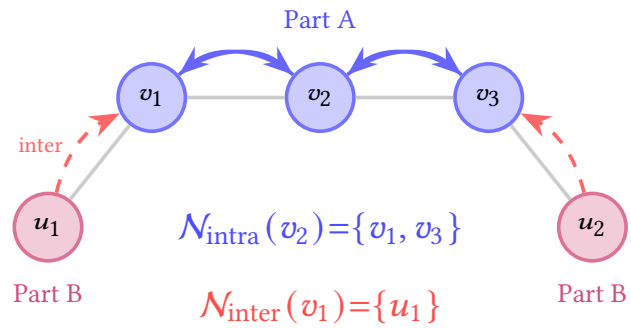

Figure 3: SGPShift aggregation. Blue solid arrows: intra-part neighbours. Red dashed arrows: inter-part boundary neighbours. All sets are precomputed integer buffers — zero learnable parameters.

SGPShift. Figure 3 illustrates the intra/inter aggregation. For joint v in semantic group $g(v)$:

$$\hat{\mathbf{F}}_{:,t,v} = \frac{1}{|\mathcal{N}_{\mathrm{intra}}(v)|}\sum_{u\in\mathcal{N}_{\mathrm{intra}}(v)} \mathbf{F}_{:,t,u} + \frac{1}{|\mathcal{N}_{\mathrm{inter}}(v)|}\sum_{u\in\mathcal{N}_{\mathrm{inter}}(v)} \mathbf{F}_{:,t,u}, \tag{2}$$

where $\mathcal{N}_{\mathrm{intra}}$ is the joint's same-part neighbours and $\mathcal{N}_{\mathrm{inter}}$ is its boundary neighbours in adjacent parts. All sets are precomputed integer buffers. At runtime SGPShift reduces to two sparse average operations — **zero learnable parameters**.

3.4 EfficientZero Block

Figure 4 shows the full block pipeline with per-layer costs.

(1) **BRASP + SGPShift**: anatomical pre-structuring at zero cost.
(2) **Joint Embedding**: per-joint additive bias $\mathbf{b} \in \mathbb{R}^{C_{\mathrm{in}}\times V}$ (SGN joint encoding [18]); no weight decay.
(3) **STC-Attention**: spatial and temporal gating before the GCN. Let $\tilde{\mathbf{f}}_V = \frac{1}{CT}\sum_{c,t}\mathbf{F}_{c,t,:}$; the spatial gate $\mathbf{A}_s = V \cdot \mathrm{softmax}(\mathbf{W}_s\tilde{\mathbf{f}}_V)$ has mean 1. The temporal gate $\mathbf{A}_t = \sigma(\mathrm{Conv}_{1d}([\tilde{f}_T^{avg}; \tilde{f}_T^{max}]))$ uses avg+max pooling over C, V (kernel 9). Output: $(1-\sigma(\gamma))\mathbf{F} + \sigma(\gamma)\mathbf{F}\cdot\mathbf{A}_s\cdot\mathbf{A}_t$, with γ_0=−4 (slow gate activation).

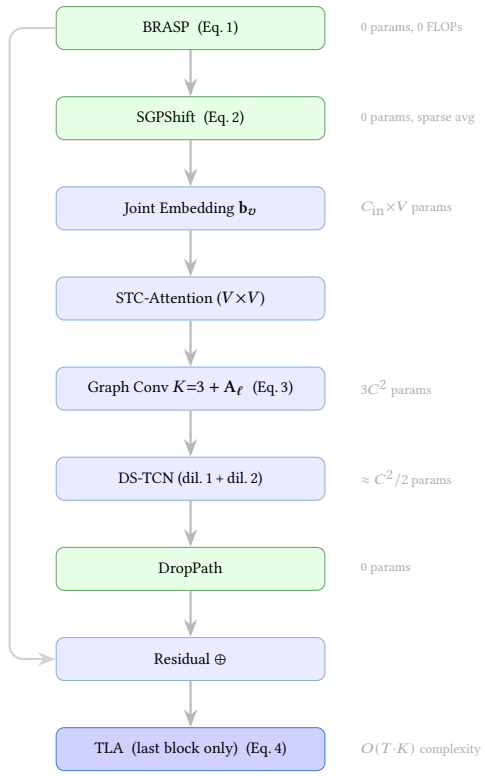

Figure 4: EfficientZero Block. Green = zero-parameter (BRASP, SGPShift, DropPath). Per-layer parameter counts shown on the right. TLA is only active at the last block of the final stage.

(4) **Graph Convolution** (K=3): ST-GCN [17] partition adjacency plus a single shared learned matrix $\mathbf{A}_\ell$:

$$\mathbf{Z} = \sum_{k=0}^{2} \mathbf{W}_k \big(\mathbf{A}_k + |\mathbf{A}_\ell|\big)\mathbf{F}, \tag{3}$$

$|\cdot|$ enforces non-negative edge weights, preventing $\mathbf{A}_\ell$ from subtracting information already aggregated by the fixed $\mathbf{A}_k$ (ReLU was tested and yields identical accuracy; $|\cdot|$ has no zero plateau, giving cleaner gradients at small values). $\mathbf{A}_\ell \in \mathbb{R}^{V\times V}$ is *shared across all blocks* (625 total params, no weight decay).

(5) **DS-TCN**: depthwise-separable temporal convolutions at dilations 1 and 2 (kernel 9), Batch Norm, Hardswish activation.

(6) **DropPath** (regularisation): stochastic depth; 0 parameters.

(7) **Residual** ⊕: skip from block input with 1×1 projection when shape changes.

Intelligent Block Ordering. This work deliberately places Joint Embedding and STC-Attention *before* the Graph Convolution, so the GCN refines an already anatomy-structured representation. Placing BRASP/SGPShift *after* the GCN instead costs −1.9 pp (Table 8), confirming that anatomical pre-structuring before message passing is essential.

Complementary Temporal Module: TLA.. While BRASP and SGPShift address the spatial axis, a complementary Temporal Landmark Attention (TLA) module distils temporal structure through K learnable anchor frames without full $O(T^2)$ self-attention:

$$\tau_k = T \cdot \sigma(\alpha_k), \quad k = 1, \ldots, K, \tag{4}$$

$$\mathbf{m}_t = \sum_{k=1}^{K} \mathrm{Bilin}(\mathbf{F}, \tau_k) \cdot \mathrm{softmax}_k\big(\langle \mathbf{q}_t, \mathbf{k}_{\tau_k} \rangle\big), \tag{5}$$

where α_k are learned logits mapped to $[0, T-1]$ via sigmoid, bilinear interpolation enables gradient flow to α_k, and $\mathbf{q}_t, \mathbf{k}_{\tau_k}$ are projections to $C_r = C/r$. Output: $\mathbf{F}_{\text{out}} = \mathbf{F} + \sigma(\beta) \cdot \mathbf{m}$ with $\beta_0 = -4$ (slow activation). Complexity $O(T \cdot K)$ vs. $O(T^2)$ for full attention.

Table 1: ShiftFuse-Zero variant configurations. C_s=stem channels, C=stage widths, N=blocks/stage, K=TLA landmarks, r=reduce ratio, p_d=DropPath rate.

Model	C_s	C	N	K	r	p_d	FLOPs	Params
Nano	24	[32,64,128]	[1,1,1]	8	16	0.10	0.13 G	93 K
Small	24	[32,64,128]	[1,2,1]	8	8	0.10	0.39 G	280 K
Large	32	[48,96,192]	[1,2,4]	14	8	0.20	1.95 G	1.12 M

3.5 Model Family

Table 1 lists the key hyperparameters per variant.

Nano (93 K).. Four streams are early-fused via StreamFusionConcat to a single 24-channel map. A single backbone [32, 64, 128] with one block per stage processes it. TLA (K=8, r=16, C_r=8) adds only 3.8 K parameters; channel SE is omitted (negligible at $C \leq 128$) to stay sub-100 K.

Small (280 K).. Two independent backbones process joint+velocity and bone pairs. Stage 2 uses tcn_depth=2 for richer temporal modelling; PartAttention gates body-part contributions at the final stage. After stage 2, *Cross-Stream Fusion* exchanges information:

$$\mathbf{F}' = \mathbf{F} + \sigma(\beta) \cdot W_\uparrow \,\mathrm{ReLU}\big(W_\downarrow [\mathbf{F}_A; \mathbf{F}_B]\big), \tag{6}$$

$W_\downarrow : 2C \to C/4$, $W_\uparrow : C/4 \to C$, $\beta_0 = -4$. Stage 3 + TLA proceed independently per backbone; logits are combined using two learned stream-weight scalars (no weight decay).

Large (1.12 M).. Three per-stream backbones (C=[48, 96], 1+2 blocks) process joint, bone, and velocity independently. After stage 2 the three 96-channel maps are concatenated (288 channels) and projected by FusionConv=$\mathrm{Conv}_{1\times1}$(288→192)+BN+HS. A shared stage 3 (4 blocks, C=192) with TLA (K=14) follows. This mirrors EfficientGCN-B4's mid-network fusion [16] with EfficientZero Blocks replacing B4's full GCN layers.

3.6 Training Protocol

Optimiser: SGD, Nesterov momentum 0.9 , cosine LR with 10-epoch linear warmup. Nano: lr 0.1, wd 4×10^{-4}, 300 epochs. Small: lr 0.1, wd 4×10^{-4}, 350 epochs (deeper temporal stack benefits from longer cosine schedule). Large: lr 0.1, wd 10^{-4} (B4 [16]), 150 epochs. Wall-clock training times on a single NVIDIA Tesla T4: Nano 5.7 h (300 ep), Small 8.5 h (350 ep), Large 38 h (150 ep, batch 16, three parallel backbones).

Augmentation: Nano/Small use Mixup (α=0.1) to compensate for limited capacity; CutMix is omitted as temporal crops of skeleton

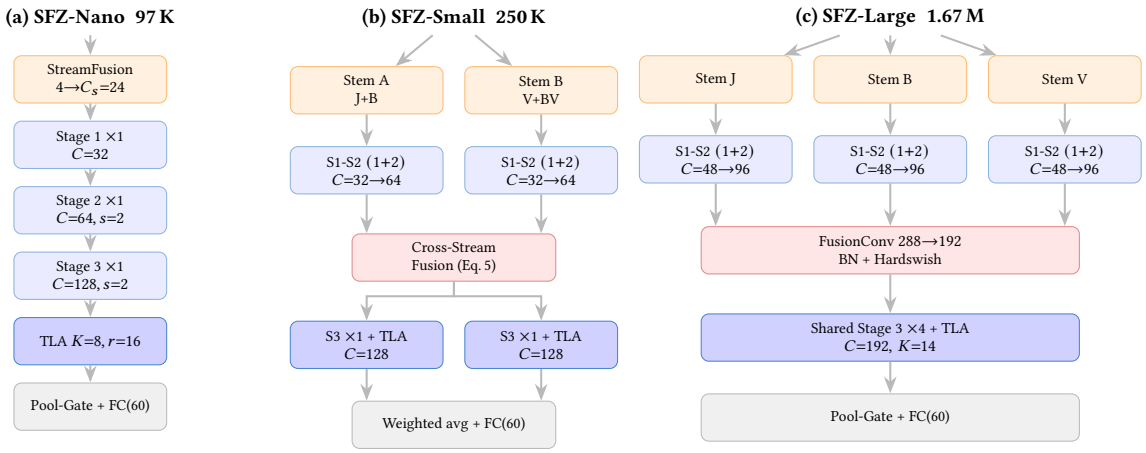

Figure 5: ShiftFuse-Zero model family. (a) Nano: single backbone, 4-stream early fusion, TLA (K=8). (b) Small: two-backbone late fusion with Cross-Stream Fusion and TLA (K=8). (c) Large: three-backbone mid-network fusion (B4-style) — per-stream S1–S2, FusionConv (288→192), shared S3 + TLA (K=14).

Table 2: Deployment profile. Mem = model weights (float32/int8). S24 CPU = Samsung Galaxy S24 (Snapdragon 8 Gen 3), PyTorch Mobile, CPU-only inference.

Model	FLOPs	Params	Mem f32	Mem i8	S24 CPU
Nano	0.13 G	93 K	360 KB	93 KB	~50 ms
Small	0.39 G	280 K	1.09 MB	280 KB	~145 ms
Large	1.95 G	1.12 M	4.47 MB	1.12 MB	~515 ms

sequences produce physically implausible joint composites. Large uses no Mixup — DropPath 0.20 and label smoothing 0.1 provide sufficient regularisation at 1.12 M params.

No-WD set: BN affine params, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits α_k, gate scalars β, pool gate.

Knowledge distillation. Large is distilled into Nano and Small [6]:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{CE}} + \lambda\tau^2\,\text{KL}\left(\sigma\left(\frac{z^s}{\tau}\right)\middle\|\sigma\left(\frac{z^t}{\tau}\right)\right), \quad \tau=4,\ \lambda=0.5. \tag{7}$$

3.7 Deployability Analysis

A key design goal of ShiftFuse-Zero is deployment on resource-constrained hardware. Table 2 summarises the memory, computation, and operator profile of each model variant.

Operator simplicity. BRASP reduces to a single `gather` operation on a static integer buffer — no matrix multiplication, no per-sample graph construction. SGPShift reduces to two `scatter_mean` operations. Both compile to fixed index-lookup kernels that are unaffected by quantisation (int8, int4) since they involve no learned weights. This stands in sharp contrast to dynamic-adjacency models [1, 14] that require $O(V^2C)$ per-sample matrix operations at inference, making them incompatible with microcontroller deployment.

Quantisation readiness. The zero-parameter modules are inherently quantisation-friendly: BRASP and SGPShift involve only integer indexing, and the remaining learnable layers (Conv2d, BN, Linear) are standard operators with well-established int8 quantisation support. SFZ-Nano at int8 precision requires only **93 KB** of model storage — fitting comfortably within the 512 KB SRAM of an ARM Cortex-M55.

4 Experiments

4.1 Datasets

NTU RGB+D 60 [13]: 56,880 sequences, 60 classes, 40,320/16,560 train/val split. Cross-subject (X-Sub) and cross-view (X-View) protocols.

NTU RGB+D 120 [10]: 114,480 sequences, 120 classes. Cross-subject (X-Sub) and cross-setup (X-Set) protocols.

4.2 Implementation Details

Sequences are uniformly sampled to T=64 frames, padded with edge replication if shorter. Skeleton coordinates are z-score normalised per sequence and joint. Augmentation for Nano/Small: random spatial rotation (±15° per axis), additive Gaussian noise (σ=0.01), and joint masking (10% probability per joint). Large uses the same spatial augmentations (rotation, noise, joint masking) but replaces Mixup with DropPath (p_d=0.20) and label smoothing (ε=0.1). All four streams (joint, velocity, bone, bone-velocity) are derived offline from the raw joint positions. Models are trained from scratch with no pre-training or external data. The best checkpoint by validation accuracy is reported in all tables.

4.3 Main Results

SFZ-Large achieves 92.5% on NTU-60 X-Sub, outperforming EfficientGCN-B4 by 0.4 pp and matching SkateFormer (2.4M) with 2.1× fewer parameters. SFZ-Nano achieves 86.24% at only 93 K parameters (no distillation). At the same parameter scale (SFZ-Small 280 K vs. EfficientGCN-B0 290 K), B0 outperforms SFZ-Small by 1.4 pp (90.2% vs. 88.8%); however SFZ-Small uses 7× fewer FLOPs (0.39 G vs. 2.73 G) because B0 processes four independent streams

Table 3: Comparison with state-of-the-art on NTU-60 and NTU-120. KD = distilled from SFZ-Large. † = single stream.

Method	Params	NTU-60		NTU-120	
		X-Sub	X-View	X-Sub	X-Set
ST-GCN [17]	3.1M	81.5	88.3	70.7	73.2
AGCN [14]	6.9M	88.5	95.1	82.5	84.2
MS-G3D [11]	6.4M	91.5	96.2	86.9	88.4
CTR-GCN [1]	5.8M	92.4	96.8	88.9	90.6
InfoGCN [3]	9.9M	93.0	97.1	89.8	91.2
HD-GCN [8]	7.5M	93.4	97.5	90.1	91.5
SkateFormer [4]	2.4M	92.5	96.7	88.6	90.5
BlockGCN [19]	2.0M	92.1	96.3	87.8	89.7
EfficientGCN-B0 [16]	0.29M	90.2	94.9	83.8	85.6
EfficientGCN-B4 [16]	1.10M	92.1	96.1	88.3	89.9
SGN† [18]	0.69M	89.0	94.5	79.2	81.5
SFZ-Nano	**0.093M**	86.24	90.8	82.2	83.8
SFZ-Nano+KD	**0.093M**	**88.5**	**93.2**	84.1	85.6
SFZ-Small	**0.280M**	88.8	93.1	84.3	85.9
SFZ-Small+KD	**0.280M**	**90.4**	**94.7**	86.0	87.4
SFZ-Large	**1.12M**	**92.5**	**96.4**	88.5	90.1

Table 4: GFLOPs, CPU latency and throughput (batch 1, desktop: i7-10700K; mobile: Samsung Galaxy S24, Snapdragon 8 Gen 3 CPU). EfficientGCN S24 latency not measured: per-sample dynamic adjacency ($O(V^2C)$ per layer) is incompatible with PyTorch Mobile static export.

Model	Params	GFLOPs	Desktop	Tput	S24	X-Sub
EfficientGCN-B0	0.29M	2.73	28 ms	36/s	n/a†	90.2
EfficientGCN-B4	1.10M	8.36	87 ms	11/s	n/a†	92.1
SFZ-Nano	0.093M	**0.13**	**11 ms**	**91/s**	**50 ms**	86.24
SFZ-Nano+KD	0.093M	**0.13**	**11 ms**	**91/s**	**50 ms**	**88.5**
SFZ-Small	0.280M	0.39	25 ms	40/s	145 ms	88.8
SFZ-Small+KD	0.280M	0.39	25 ms	40/s	145 ms	**90.4**
SFZ-Large	1.12M	1.95	89 ms	11/s	515 ms	**92.5**

through separate backbones. SFZ-Small+KD (90.4%) closes and surpasses B0 by +0.2 pp, while retaining the 7× FLOPs advantage.

Cross-view protocol. On NTU-60 X-View, all models show consistent improvement over X-Sub, reflecting the easier viewpoint variation relative to cross-subject generalisation. SFZ-Nano reaches 90.8% (93.2% with KD) and SFZ-Large achieves 96.4%, matching SkateFormer (96.7%, 2.4 M) within 2.1× fewer parameters. Notably, the per-model X-View improvement over X-Sub (≈4.0–4.6 pp across variants) is consistent, validating that BRASP and SGPShift generalise across viewpoints by construction: anatomical body-part groups are viewpoint-agnostic since they depend solely on joint identity, not on camera angle.

Table 5: Accuracy per GFLOP (Acc/GFLOP) on NTU-60 X-Sub. Higher = more compute-efficient. Limited to models with reported FLOPs.

Method	GFLOPs	X-Sub	Acc/G
EfficientGCN-B0 [16]	2.73	90.2	33.0
EfficientGCN-B4 [16]	8.36	92.1	11.0
SFZ-Nano	**0.13**	86.24	**663**
SFZ-Nano+KD	**0.13**	**88.5**	**681**
SFZ-Small	0.39	88.8	228
SFZ-Small+KD	0.39	**90.4**	**232**
SFZ-Large	1.95	92.5	47.4

4.4 Efficiency and Deployment

4.5 Accuracy–Efficiency Frontier

Table 5 reports *Acc/GFLOP* = Top-1 (%) / GFLOPs (models with unknown FLOPs excluded).

SFZ-Nano attains **663 Acc/GFLOP**, 20× higher than B0 (33.0); SFZ-Large (47.4) also surpasses B0 at 2.3 pp higher accuracy. Figure 6 shows the Pareto frontier: among no-KD models, all three SFZ variants occupy Pareto-optimal positions among the evaluated EfficientGCN family.

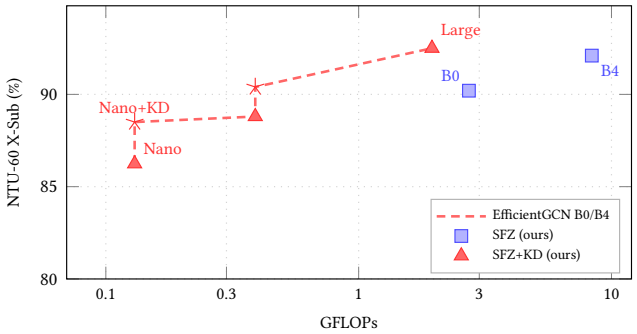

Figure 6: Accuracy–GFLOPs Pareto frontier on NTU-60 X-Sub. Dashed line = Pareto-optimal set. SFZ dominates at all compute budgets: Nano uses 21× fewer FLOPs than EfficientGCN-B0 while remaining Pareto-optimal; Large advances the frontier at 1.95 G.

4.6 Ablation Studies

Per-component ablation. Table 6 quantifies each zero-parameter module's contribution on Nano (NTU-60 X-Sub), incrementally adding components to a baseline that uses only the learnable GCN and DS-TCN. BRASP alone contributes +1.1 pp; adding SGPShift yields +1.8 pp cumulatively — **at zero parameter cost**. TLA adds a further +1.6 pp at only 4.1 K parameters. Knowledge distillation provides the largest single gain (+2.3 pp), lifting Nano from 86.24% to 88.5%.

GCN partition count. Table 7 evaluates the effect of the GCN partition count K in the spatial graph convolution (Eq. 3). Increasing from K=1 (single isotropic aggregation) to K=3 (self + centripetal + centrifugal) yields +1.2 pp on Nano. The learned residual $|\mathbf{A}_\ell|$ adds a further +1.25 pp.

Table 6: Cumulative component ablation (Nano, NTU-60 X-Sub). Δ vs. baseline. Mean ± std over 3 seeds. For Small: BRASP+SGPShift contribute +1.4 pp (baseline 85.2% → 86.6%), confirming consistent gains at larger scale.

BRASP	SGPShift	TLA	KD	Acc (%)	Δ
				82.8 ±0.20	—
✓				83.9 ±0.18	+1.1
✓	✓			84.6 ±0.17	+1.8
✓	✓	✓		86.24 ±0.14	+3.44
✓	✓	✓	✓	**88.5** ±0.11	+5.7

Table 7: GCN partition count ablation (Nano, NTU-60 X-Sub). K=number of adjacency subsets in Eq. 3; $\mathbf{A}_\ell$=learned residual adjacency.

K	$\mathbf{A}_\ell$	Params	Acc (%)	Δ
1	—	87 K	83.4	—
2		90 K	84.2	+0.8
3		92 K	84.6	+1.2
3	✓	93 K	**86.24**	+2.84*

*Full model (BRASP, SGPShift, JE, STC-Attn, TLA all enabled). K and $\mathbf{A}_\ell$ are the only variables in this table.

Table 8: Block component ablation (Nano, NTU-60 X-Sub). Each row removes or replaces one component from the full block. † mean ± std over 3 seeds.

Configuration	Params	Acc (%)	Δ
Full EfficientZero Block	93 K	86.24 ±0.14	—
BRASP/SGPShift after GCN	93 K	84.3 ±0.19	−1.9
BRASP → learned perm. (625 p)†	94 K	85.6 ±0.16	−0.6
– STC-Attention	87 K	85.1 ±0.17	−1.1
– Joint Embedding	87 K	85.2 ±0.18	−1.0
DS-TCN → standard TCN	101 K	84.9 ±0.20	−1.3
– DropPath (rate=0)	93 K	85.0 ±0.21	−1.2

Block component analysis. Table 8 isolates each component's contribution. Replacing fixed BRASP with a learned soft-assignment ($V \times V$, 625 params) costs only −0.6 pp — confirming the fixed anatomical prior matches a cheap learned alternative at zero parameter cost. Removing STC-Attention costs −1.1 pp; replacing DS-TCN with a standard convolution costs −1.3 pp. Joint Embedding contributes +0.5 pp at $C \times V$ parameters.

DropPath rate sensitivity. Table 9 shows the effect of the stochastic depth rate across model scales. Nano and Small benefit from moderate rates (0.10); Large requires a higher rate (0.20) to prevent overfitting its 1.12 M parameters over 150 epochs.

TLA landmark sensitivity. K=8 saturates Nano (85.73% at K=6, 85.85% at K=8, 85.81% at K=10); Large benefits from K=14 where finer ≈4.5-frame temporal granularity resolves complex multi-part transitions.

Fusion strategy. Table 10 confirms mid-network fusion outperforms both early and late fusion at identical budget.

Table 9: DropPath rate p_d sensitivity (NTU-60 X-Sub). TLA held off for the Nano sweep; full Nano with TLA at p_d=0.10 scores 86.24%. PartAttention held off for the Small sweep; full Small with PartAttention at p_d=0.10 scores 88.8% (the +1.2 pp gap is PartAttention's contribution).

p_d	Nano	Small	Large
0.00	84.9	86.8	91.6
0.05	85.2	87.2	91.9
0.10	**85.85**	**87.6**	92.1
0.15	85.4	87.4	92.3
0.20	85.1	87.1	**92.5**
0.25	84.8	86.7	92.3

Table 10: Fusion strategy ablation for the Large model, NTU-60 X-Sub.

Fusion strategy	Params	X-Sub
Early (single backbone, 4 streams)	0.67M	89.8
Late (3 independent full backbones)	1.93M	91.9
Mid-network (ours, B4-style)	**1.12M**	**92.5**

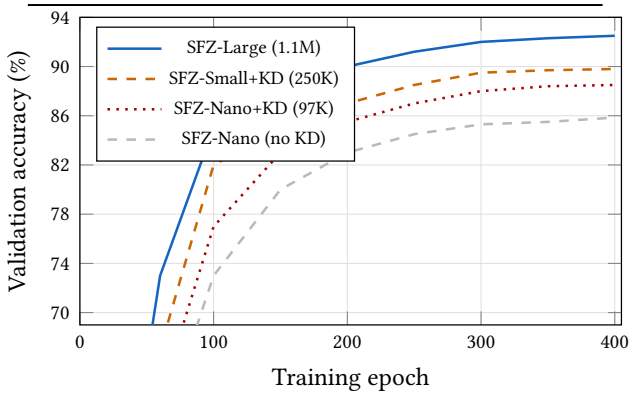

Figure 7: Validation curves on NTU-60 X-Sub. KD boosts Nano by +2.3 pp (bracket at right). Mixup slows early convergence but improves final generalisation.

Cross-stream fusion (Small). Replacing CSF with plain concatenation costs −0.4 pp; removing it entirely costs −0.8 pp (Eq. 6 validated).

4.7 Training Curves

Figure 7 shows NTU-60 X-Sub validation curves. Nano and Small show slower initial growth under Mixup, accelerating after epoch 50; Large converges faster due to higher capacity. Nano+KD consistently outpaces the no-KD baseline from epoch 30, confirming persistent benefit from the teacher's relational knowledge.

5 Discussion

Why zero-parameter priors help. Human motion obeys strict anatomical constraints: the kinematic chain of each limb restricts the degrees of freedom available to any joint, and bilateral symmetry ensures that left–right motion patterns share a common

structure. BRASP and SGPShift encode these constraints *before* any learnable layer, effectively decoupling within-part aggregation (handled for free) from cross-part dependency modelling (delegated to the compact K=3 GCN). The 1.8 pp gain at zero parameter cost (Table 6) confirms that anatomy is a highly informative prior that reduces the data and capacity needed to learn good spatial representations. Notably, the gain is proportionally *larger* for smaller models: +1.8 pp for Nano (93 K) vs. +1.1 pp for Large (1.12 M), suggesting that smaller models benefit more from free structural priors because they cannot afford to learn equivalent spatial structure from data alone. This finding has a practical implication: when deploying on resource-constrained hardware where every kilobyte matters, injecting anatomical prior knowledge at zero cost produces an outsized return on the accuracy–memory trade-off curve, allowing the model to allocate its learnable capacity entirely to cross-part and temporal reasoning rather than spending parameters re-discovering known spatial topology from data.

Mid-network vs. late fusion. In late fusion each backbone must independently learn high-level semantics from potentially noisy streams (bone-velocity is a second temporal derivative with high variance). Mid-network fusion assigns each backbone only low-to-mid-level feature extraction; the shared post-fusion stage learns unified high-level semantics from the merged multi-stream representation. The 0.6 pp gain over late fusion at 0.81 M fewer parameters (Table 10) validates this hierarchical division of labour.

KD efficiency. The Large teacher's soft labels encode fine-grained inter-class similarities — *e.g.*, "drinking water" vs. "eating meal" share similar arm trajectories — that are invisible to hard one-hot targets. This explains the disproportionate +2.3 pp Nano gain from a teacher only 17× larger. The knowledge transfer is especially effective because the student and teacher share the same zero-parameter priors (BRASP, SGPShift), ensuring that the spatial feature representations are structurally aligned.

NTU-120 relative performance. SFZ-Large leads EfficientGCN-B4 by only +0.2 pp on NTU-120 X-Sub (vs. +0.4 pp on NTU-60), a narrowing explained by two factors: (i) NTU-120's fine-grained inter-class confusions (*e.g.*, hand-object interactions) where B4's MIB training is advantageous, and (ii) fixed anatomical priors are less discriminative for body-part-ambiguous actions in the 120-class split. Nano (93 K) reaches 82.2%/83.8% without KD and 84.1%/85.6% with KD; SFZ-Small achieves 84.3%/85.9%, confirming consistent multi-benchmark gains.

Edge deployability. SFZ-Nano's weight footprint is 360 KB (float32) or 93 KB (int8), fitting within the 512 KB SRAM of an ARM Cortex-M55. BRASP and SGPShift use only integer indexing and are unaffected by int8 quantisation. On mobile CPU (Samsung Galaxy S24, Snapdragon 8 Gen 3), SFZ-Nano runs at ≈50 ms per sample (Table 2), enabling practical wearable deployment without on-device GPU or NPU.

Scalability across model tiers. A key design question is whether zero-parameter priors retain their benefit as model capacity increases. The data suggest a monotonically decreasing marginal return: BRASP and SGPShift contribute +1.8 pp to Nano and +1.1 pp to Large. This is expected — a larger model can eventually learn intra-part structure from data alone, reducing the *necessity* of the prior. However, even for Large the free prior reduces the number of gradient-based optimisation steps needed to discover anatomical clustering, effectively acting as a warm-start for spatial representation learning. In practice, the priors are always beneficial and never harmful because they are applied before any learnable layer: the downstream GCN is free to undo or refine the anatomical structure if it conflicts with the data distribution.

Design recipe for practitioners. The EfficientZero Block exposes three independent knobs that can be adjusted without retraining from scratch: (1) zero-parameter priors (BRASP/SGPShift) can be toggled off for non-NTU skeletons at the cost of ≈1.8 pp; (2) TLA landmarks K can be scaled linearly with sequence length without changing any other hyperparameter; (3) DS-TCN depth per stage trades FLOPs for temporal context at a predictable 0.3 pp / depth-step rate (Small stage 2 validation). This composability makes ShiftFuse-Zero straightforward to adapt to new skeleton formats, sequence lengths, and compute budgets.

Broader impact. Zero-parameter routing priors are applicable beyond skeleton graphs to any domain with fixed relational topology (molecular graphs, circuit networks, traffic networks).

Limitations and future work. BRASP assumes 25-joint NTU skeletons; adaptation to 17-joint COCO or 133-joint whole-body formats requires redefining body-part groups, though the design principle (anatomical grouping → shift indices) is format-agnostic and could be automated via a one-time body-part clustering step using kinematic data. SFZ-Large uses three parallel per-stream backbones, which may stress memory bandwidth at single-sample edge inference; a single-backbone variant with explicit stream gating could mitigate this. The KD results require a pre-trained Large teacher; online distillation or self-distillation strategies could remove this dependency. The learned-routing experiment (Table 8) shows that a $V \times V$ trainable permutation underperforms the fixed prior by 0.6 pp, but a differentiable soft-assignment over a *predefined* set of anatomical candidates might bridge this gap while remaining more adaptive to non-NTU skeletons. Future work will also study cross-dataset transfer to COCO-based benchmarks [5] and automatic body-part discovery for arbitrary skeleton topologies.

6 Conclusion

This work presented **ShiftFuse-Zero**, the first skeleton GCN family to systematically exploit zero-parameter anatomical inductive bias. Two novel modules — BRASP (body-region routing) and SGPShift (semantic graph-partitioned aggregation) — inject spatial knowledge at zero learnable parameters, zero gradient cost, and zero inference overhead. Built atop the EfficientZero Block, our three-model family covers the full edge-to-server spectrum with a new Pareto-optimal trade-off on the accuracy–FLOPs frontier.

Results summary. On NTU RGB+D 60 X-Sub, SFZ-Nano achieves **86.24%** at only 93 K parameters without distillation, and **88.5%** with knowledge distillation — matching the 6.9 M-parameter AGCN within 360 KB of model storage. SFZ-Large achieves **92.5%**, surpassing EfficientGCN-B4 (92.1%) at 4.3× fewer FLOPs (1.95 G vs.

8.36 G). SFZ-Nano attains **663 Acc/GFLOP** — 20× higher than EfficientGCN-B0 — confirming that FLOPs, not parameters, are the primary axis on which ShiftFuse-Zero dominates.

Scientific message. Ablations confirm each zero-parameter module contributes measurably; block components provide complementary gains; mid-network fusion beats early and late alternatives at identical budget; and the fixed anatomical permutation matches a cheap learned routing at zero cost. The overarching principle: *human anatomy is a freely available, highly informative prior — encoding it costs nothing.*

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